

Pr-1

*30. A total amount of charge Q is uniformly distributed along a thin, straight plastic rod of length l .

(a) Find the electric field at the point P , at a distance x from one end of the rod (Figure 23.35).

(b) Find the electric field at point P' , at a distance y from the midpoint of the rod (see Figure 23.35).

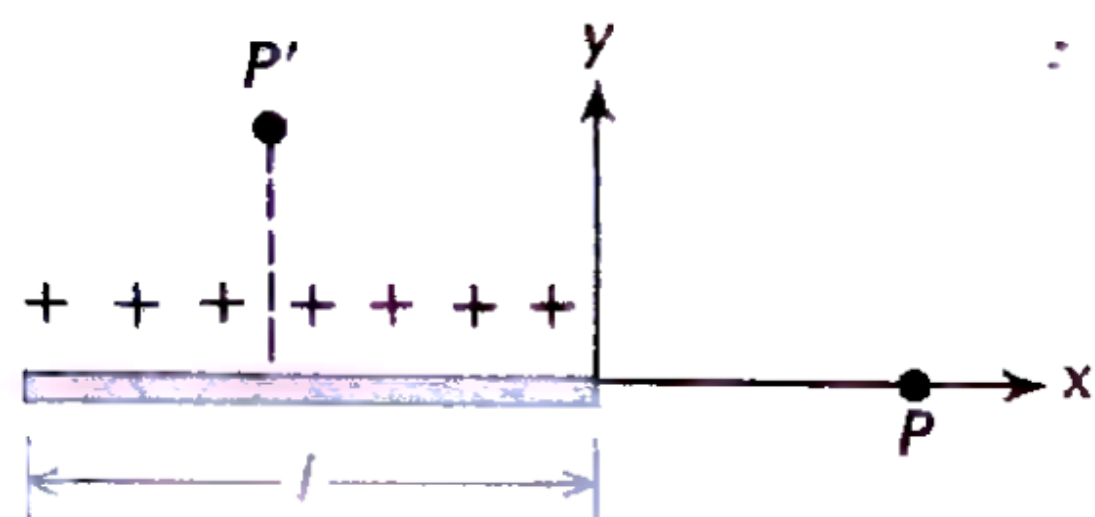


Fig. 23.35 A thin rod, with a uniform distribution of charge.

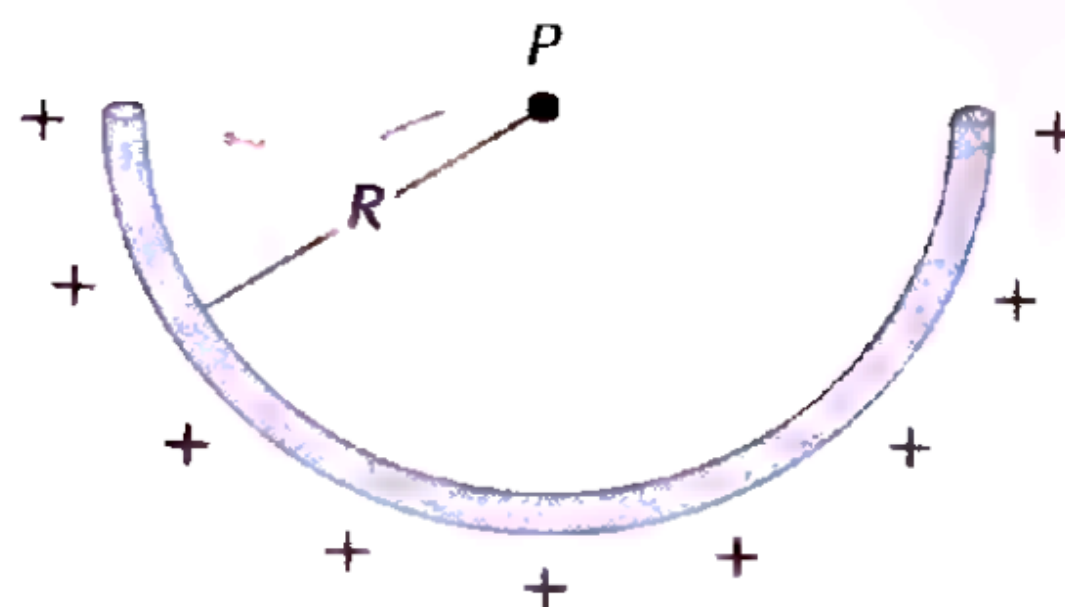


Fig. 23.36 A semicircular rod, with a uniform distribution of charge.

Problem-2

*24. Electric charge is uniformly distributed over each of three large parallel sheets of paper (Figure 23.30). The charges per unit area on the sheets are $2 \times 10^{-6} \text{ C/m}^2$, $2 \times 10^{-6} \text{ C/m}^2$, and $-2 \times 10^{-6} \text{ C/m}^2$, respectively. The distance between one sheet and the next is 1.0 cm. Find the strength of the electric field $\mathbf{E}$ above the sheets, below the sheets, and in the space between the sheets. Find the direction of $\mathbf{E}$ at each place.

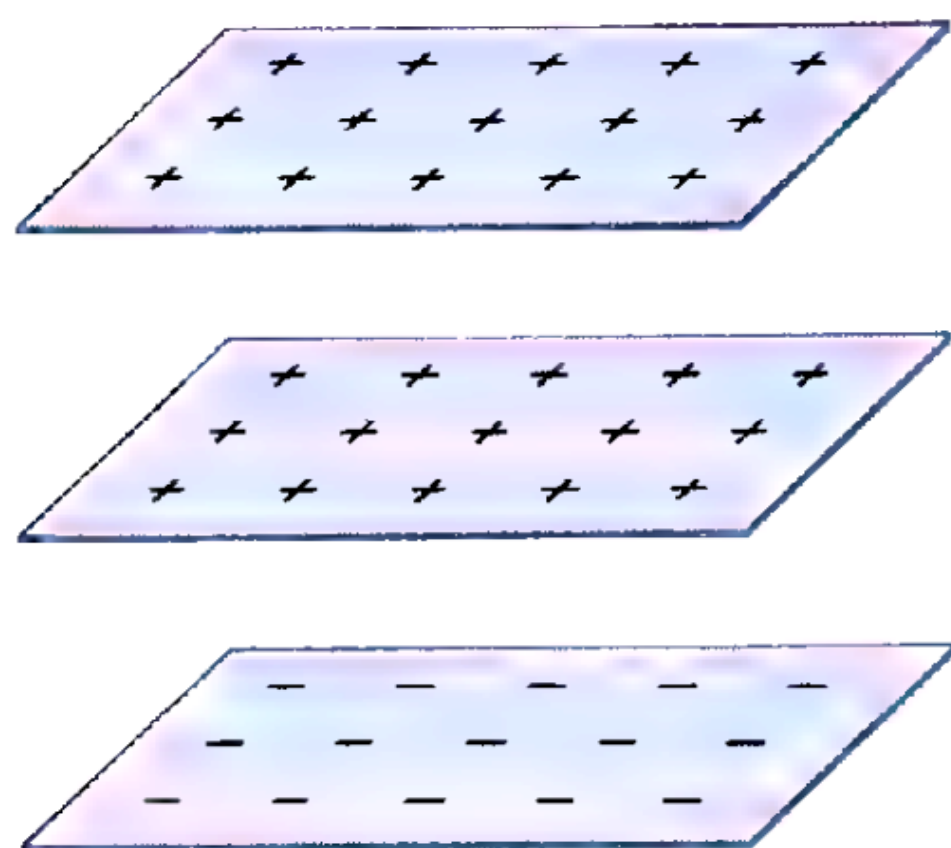
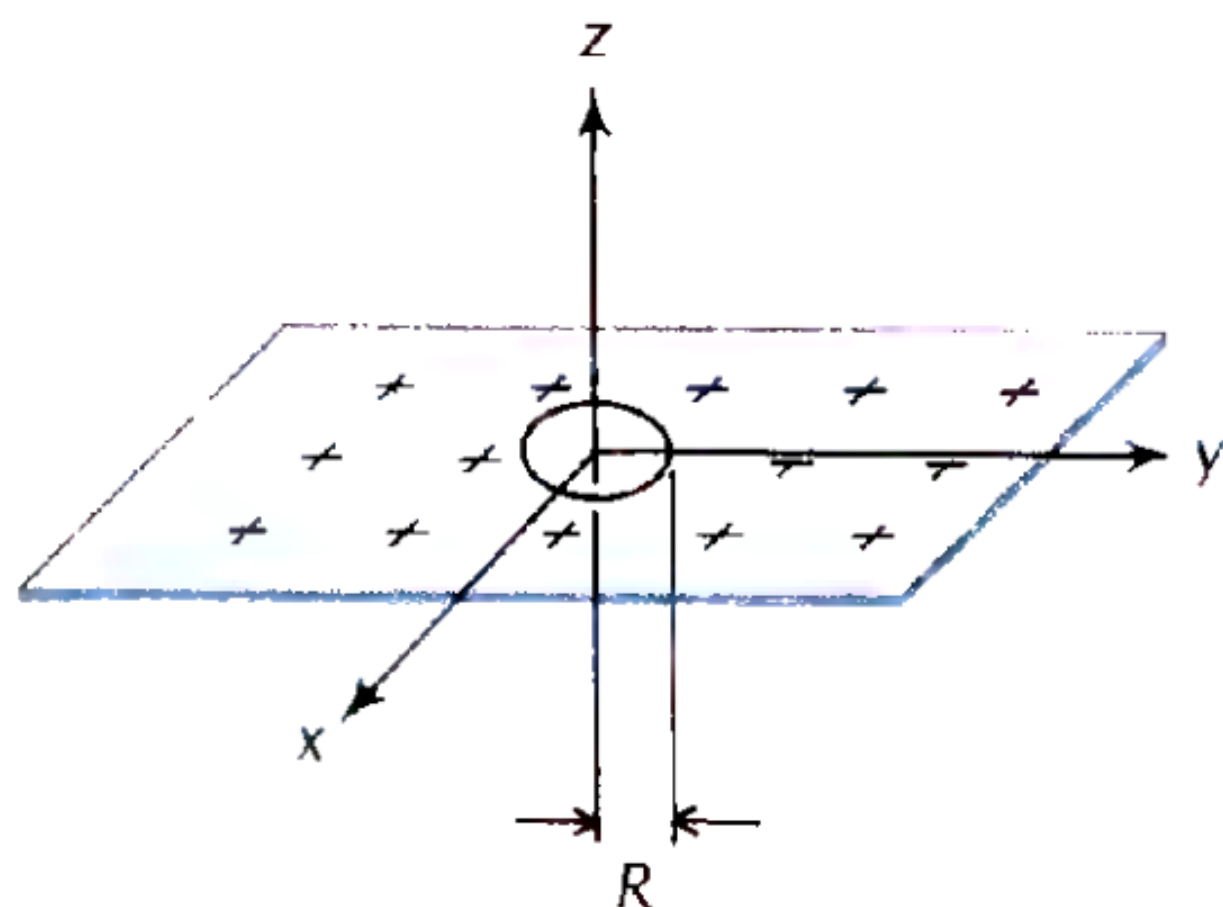


Fig. 23.30 Three parallel sheets.

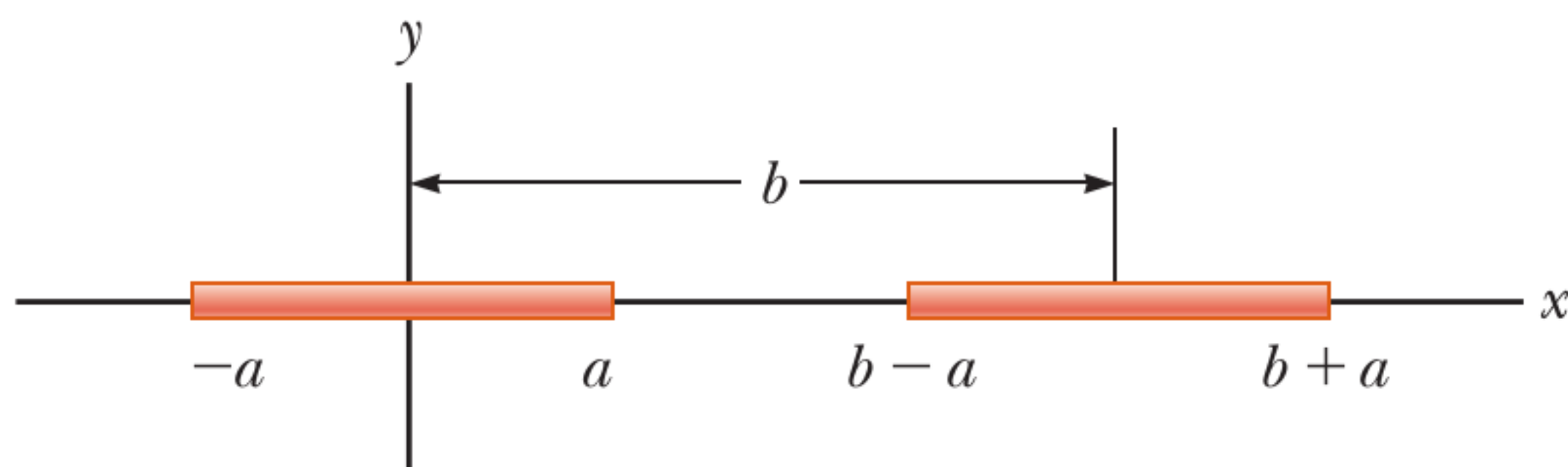
*28. A very large flat sheet of paper carries charge uniformly distributed over its surface; the amount of charge per unit area is σ . A hole of radius R has been cut out of this paper (see Figure 23.34). Find the electric field on the axis of the hole.



Challenge Problems

84. Identical thin rods of length $2a$ carry equal charges $+Q$ uniformly distributed along their lengths. The rods lie along the x axis with their centers separated by a distance $b > 2a$ (Fig. P23.84). Show that the magnitude of the force exerted by the left rod on the right one is

$$F = \left(\frac{k_e Q^2}{4a^2} \right) \ln \left(\frac{b^2}{b^2 - 4a^2} \right)$$



87. Review. An electric dipole in a uniform horizontal electric field is displaced slightly from its equilibrium position as shown in Figure P23.87, where θ is small. The separation of the charges is $2a$, and each of the two particles has mass m . (a) Assuming the dipole is released from this position, show that its angular orientation exhibits simple harmonic motion with a frequency

$$f = \frac{1}{2\pi} \sqrt{\frac{qE}{ma}}$$

What If? (b) Suppose the masses of the two charged particles in the dipole are not the same even though each particle continues to have charge q . Let the masses of the particles be m_1 and m_2 . Show that the frequency of the oscillation in this case is

$$f = \frac{1}{2\pi} \sqrt{\frac{qE(m_1 + m_2)}{2am_1m_2}}$$

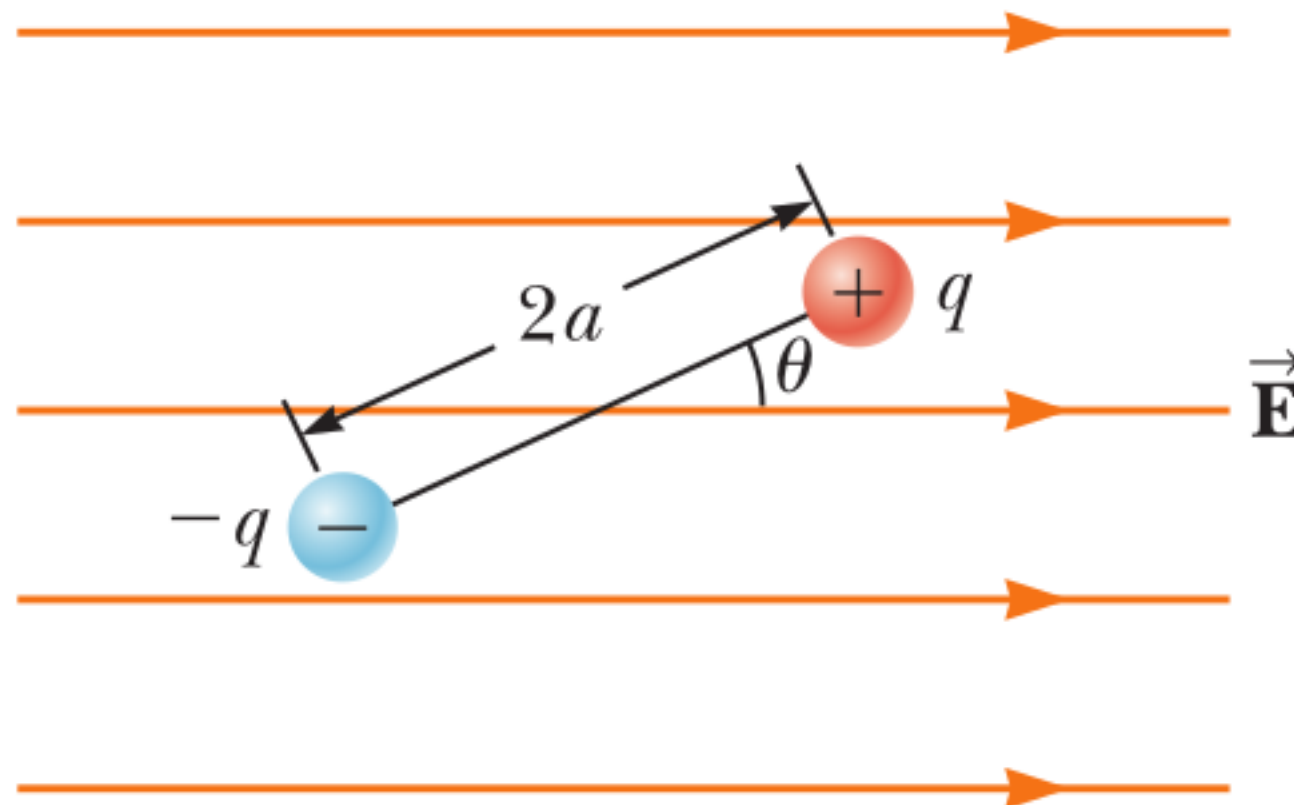


Figure P23.87